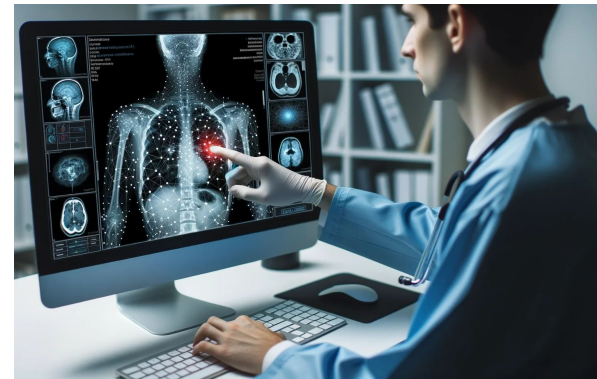
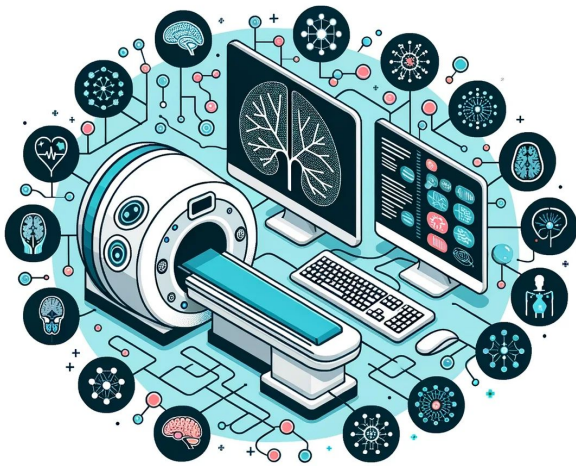


Applied Deep Learning and Generative Models in Healthcare

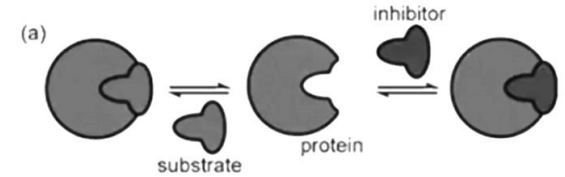
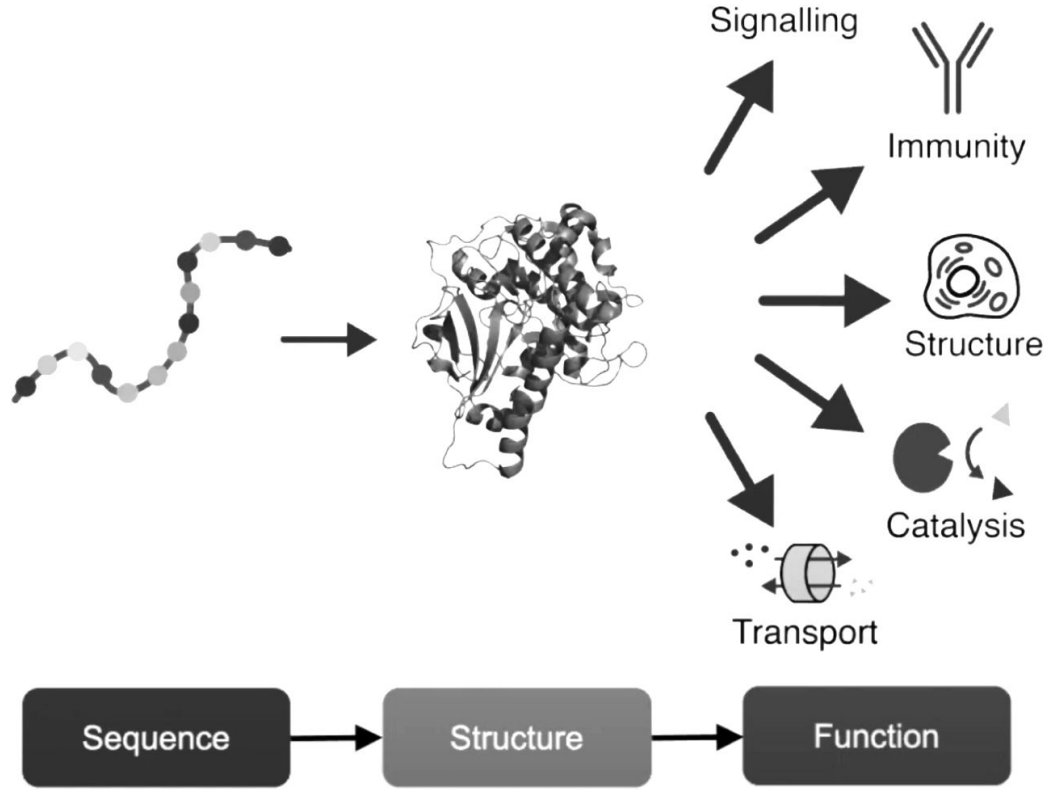


Session 12: LLM and AI in Drug Discovery

Date: April 05 2025

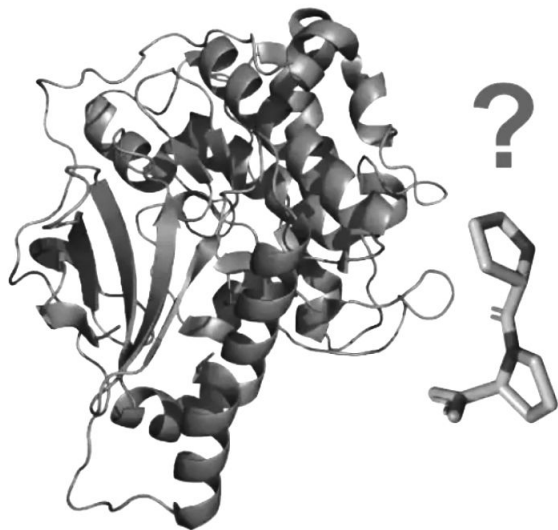
Instructor: Mahmoud E. Khani, Ph.D.

Introduction



Wilson, A. J. (2009). Chemical Society Reviews, 38(12), 3289-3300.

Introduction

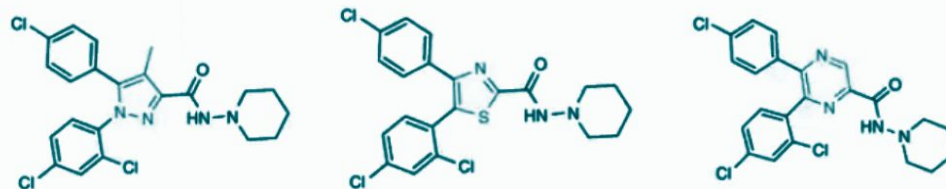


1. Fragment-based drug design [Bancet A. et al., 2020]

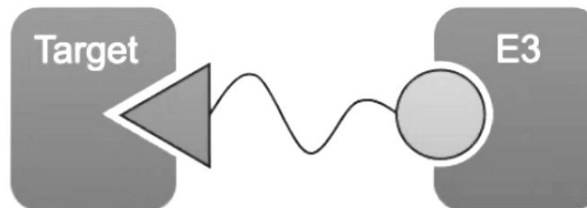


- Molecule growing
- Fragment merging
- Fragment linking

2. Scaffold hopping [Sun H. et al., 2012]



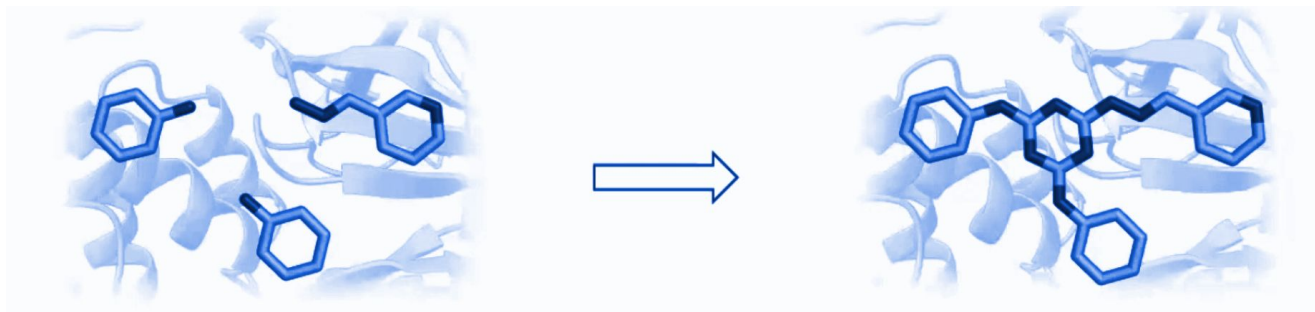
3. Targeted protein degradation [Békés M. et al., 2022]



- PROTACs
- Molecular Glues

Problem Setup

- Fragments placed in 3D space (known fixed orientation)
- Variable linker size
- Unknown attachment atoms (anchors)
- Variable number of fragments
- No clashes with protein pocket



Diffusion models

Forward diffusion process:

$$q(z_t|z_{t-1}) = \mathcal{N}(z_t; \bar{\alpha}_t z_{t-1}, \bar{\sigma}_t^2 \mathbf{I})$$

$$q(z_0, z_1, \dots, z_T|x) = q(z_0|x) \prod_{t=1}^T q(z_t|z_{t-1})$$

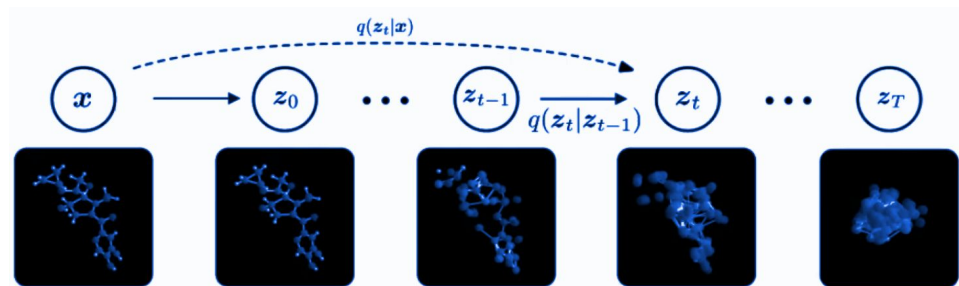
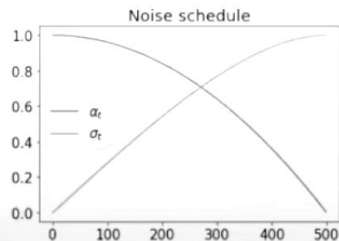
$$q(z_t|x) = \mathcal{N}(z_t|\alpha_t x, \sigma_t^2 \mathbf{I})$$

Noise parameters and schedule:

$$\bar{\alpha}_t = \alpha_t / \alpha_{t-1} \quad \alpha_t \propto 1 - (t/T)^2$$

$$\bar{\sigma}_t^2 = \sigma_t^2 - \bar{\alpha}_t^2 \sigma_{t-1}^2$$

$$\alpha_t = \sqrt{1 - \sigma_t^2}$$



- coordinates x_r
- atom types x_h

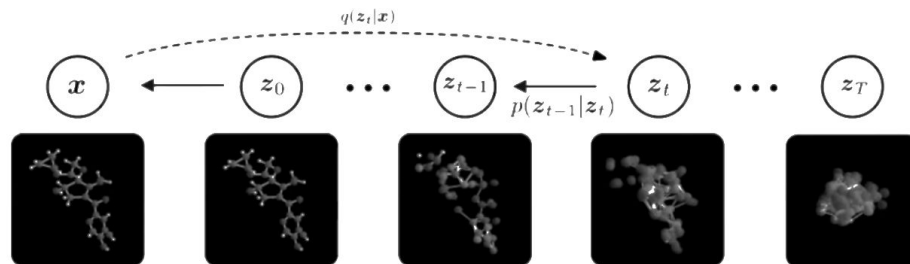
$$x = [x_r, x_h]$$

Diffusion models

Forward diffusion process:

$$q(\mathbf{z}_t | \mathbf{z}_{t-1}) = \mathcal{N}(\mathbf{z}_t; \bar{\alpha}_t \mathbf{z}_{t-1}, \bar{\sigma}_t^2 \mathbf{I})$$

$$q(\mathbf{z}_t | \mathbf{x}) = \mathcal{N}(\mathbf{z}_t | \alpha_t \mathbf{x}, \sigma_t^2 \mathbf{I})$$



True denoising process:

$$q(\mathbf{z}_{t-1} | \mathbf{x}, \mathbf{z}_t) = \mathcal{N}(\mathbf{z}_{t-1}; \boldsymbol{\mu}_t(\mathbf{x}, \mathbf{z}_t), \varsigma_t^2 \mathbf{I}), \quad \boldsymbol{\mu}_t(\mathbf{x}, \mathbf{z}_t) = \frac{\bar{\alpha}_t \sigma_{t-1}^2}{\sigma_t^2} \mathbf{z}_t + \frac{\alpha_t \bar{\sigma}_t^2}{\sigma_t^2} \mathbf{x} \quad \text{and} \quad \varsigma_t = \frac{\bar{\sigma}_t \sigma_{t-1}}{\sigma_t}$$



Generative denoising process: Instead of true $\mathbf{x}$ we use some approximation $\hat{\mathbf{x}}$

$$p(\mathbf{z}_{t-1} | \mathbf{z}_t) = q(\mathbf{z}_{t-1} | \hat{\mathbf{x}}, \mathbf{z}_t), \quad \text{where} \quad \hat{\mathbf{x}} = (1/\alpha_t) \mathbf{z}_t - (\sigma_t/\alpha_t) \hat{\boldsymbol{\epsilon}}_t$$

$$\hat{\boldsymbol{\epsilon}}_t = \varphi(\mathbf{z}_t, t)$$



Learning objective: $\mathbb{E}_{t \sim \mathcal{U}(0, \dots, T)} [T \cdot ||\boldsymbol{\epsilon} - \hat{\boldsymbol{\epsilon}}_t||^2]$

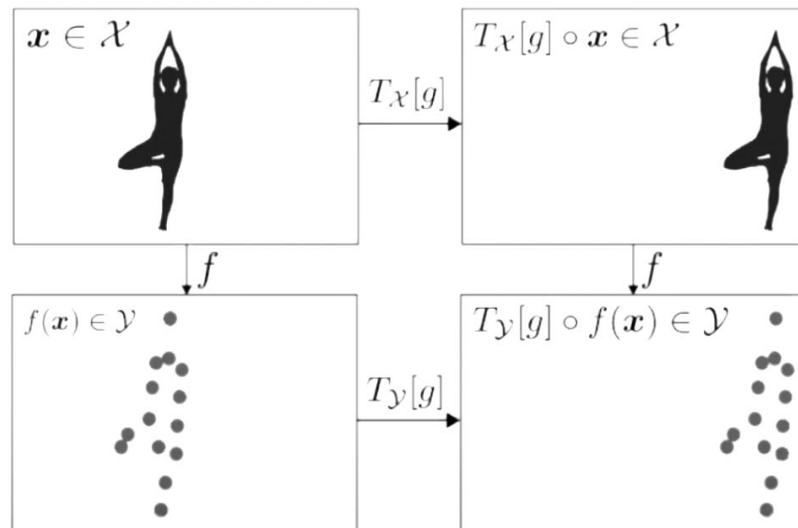
Equivariance

Function $f : \mathcal{X} \rightarrow \mathcal{Y}$ is equivariant w.r.t. group $\mathcal{G}$ if

$$f(T_{\mathcal{X}}[g] \circ x) = T_{\mathcal{Y}}[g] \circ f(x) \quad \forall g \in \mathcal{G} \quad \forall x \in \mathcal{X}$$

Distribution $p(x|y)$ is equivariant w.r.t. group $\mathcal{G}$ if

$$p(T_{\mathcal{X}}[g] \circ x | T_{\mathcal{Y}}[g] \circ y) = p(x|y) \\ \forall g \in \mathcal{G}, \forall x \in \mathcal{X}, \forall y \in \mathcal{Y}$$



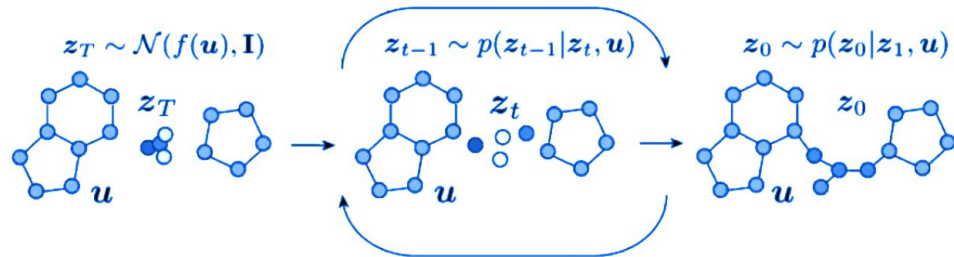
Worrall, D. E. et al. (2017) CVPR

3D Conditioning

Additional fixed 3D context $\mathbf{u}$:

$$\mathbf{x} \rightarrow [\mathbf{x}, \mathbf{u}]$$

$$\mathbf{z}_t \rightarrow [\mathbf{z}_t, \mathbf{u}]$$



$$p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{u}) = q(\mathbf{z}_{t-1}|\hat{\mathbf{x}}, \mathbf{z}_t), \quad \text{where} \quad \hat{\mathbf{x}} = (1/\alpha_t)\mathbf{z}_t - (\sigma_t/\alpha_t)\varphi(\mathbf{z}_t, \mathbf{u}, t)$$

Proposition 1 Consider a prior noise distribution $p(\mathbf{z}_T|\mathbf{u}) = \mathcal{N}(\mathbf{z}_T; f(\mathbf{u}), \mathbf{I})$ and transition distributions $p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{u}) = q(\mathbf{z}_{t-1}|\hat{\mathbf{x}}, \mathbf{z}_t)$, where q is an isotropic Gaussian, $f: \mathbb{R}^{M \times 3} \rightarrow \mathbb{R}^3$ is a function operating on 3D point clouds, and $\hat{\mathbf{x}}$ is an approximation computed by the neural network φ . Let the conditional denoising probabilistic model p be a Markov chain defined as

$$p(\mathbf{z}_0, \mathbf{z}_1, \dots, \mathbf{z}_T|\mathbf{u}) = p(\mathbf{z}_T|\mathbf{u}) \prod_{t=1}^T p(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{u}). \quad (9)$$

If f is $O(3)$ -equivariant and φ is equivariant to joint $O(3)$ -transformations of $\mathbf{z}_t$ and $\mathbf{u}$, then $p(\mathbf{z}_0|\mathbf{u})$ is $O(3)$ -equivariant.

Equivariant Graph Neural Network

- Process coordinates and features separately
- Compute interatomic distances:

$$d_{ij} = \|\mathbf{r}_i^l - \mathbf{r}_j^l\|$$

- Message:

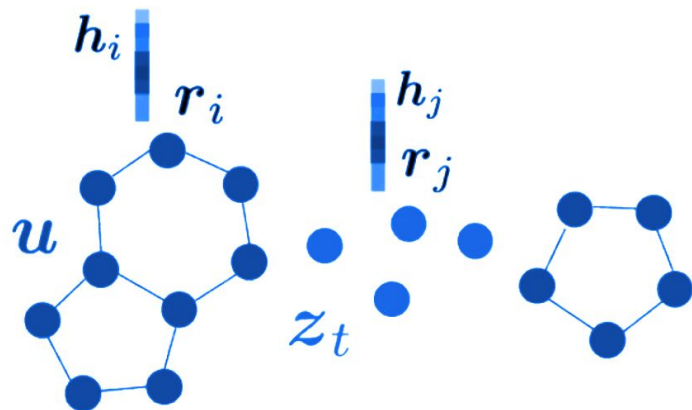
$$\mathbf{m}_{ij} = \phi_e(\mathbf{h}_i^l, \mathbf{h}_j^l, d_{ij}^2, a_{ij})$$

$$\tilde{e}_{ij} = \phi_{inf}(\mathbf{m}_{ij})$$

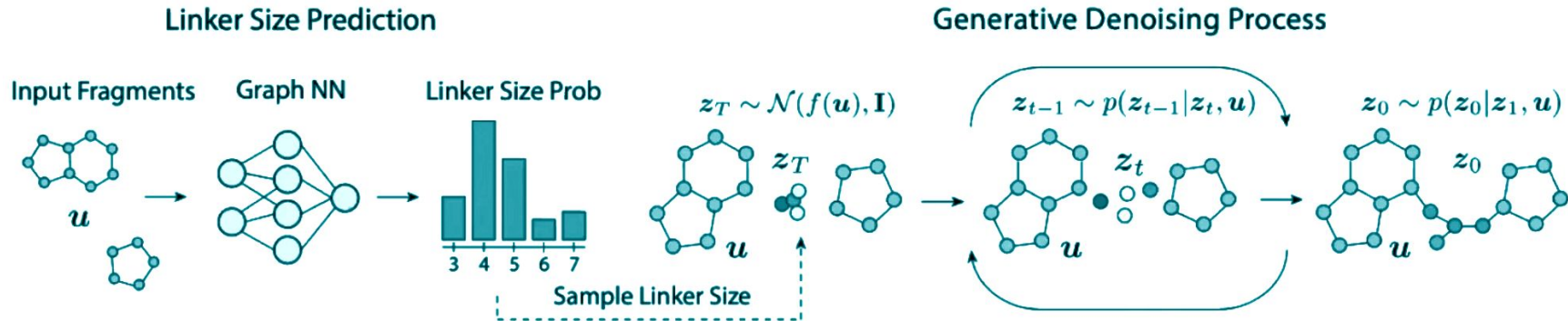
- Update:

$$\mathbf{h}_i^{l+1} = \phi_h(\mathbf{h}_i^l, \sum_{j \neq i} \tilde{e}_{ij} \mathbf{m}_{ij}) \quad \mathbf{r}_i^{l+1} = \mathbf{r}_i^l + \phi_{vel}(\mathbf{r}^l, \mathbf{h}^l, i)$$

$$\phi_{vel}(\mathbf{r}^l, \mathbf{h}^l, i) = \begin{cases} \sum_{j \neq i} \frac{\mathbf{r}_i^l - \mathbf{r}_j^l}{d_{ij} + 1} \phi_r(\mathbf{h}_i^l, \mathbf{h}_j^l, d_{ij}^2) & \text{if node } i \text{ belongs to the point cloud } \mathbf{z}_t, \\ 0 & \text{if node } i \text{ belongs to the context } \mathbf{u} \end{cases}$$



Equivariant Diffusion for Molecular Linker Design



Linker Size:

- Sampled from GNN probs
- Specified by user

Context u :

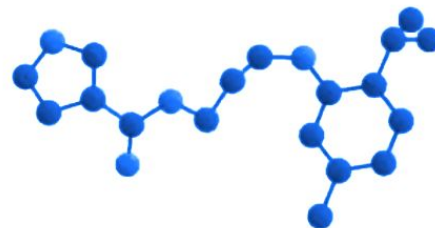
- Fragments (any number)
- Fragments + Pocket

Center f :

- CoM of anchors
- CoM of fragments

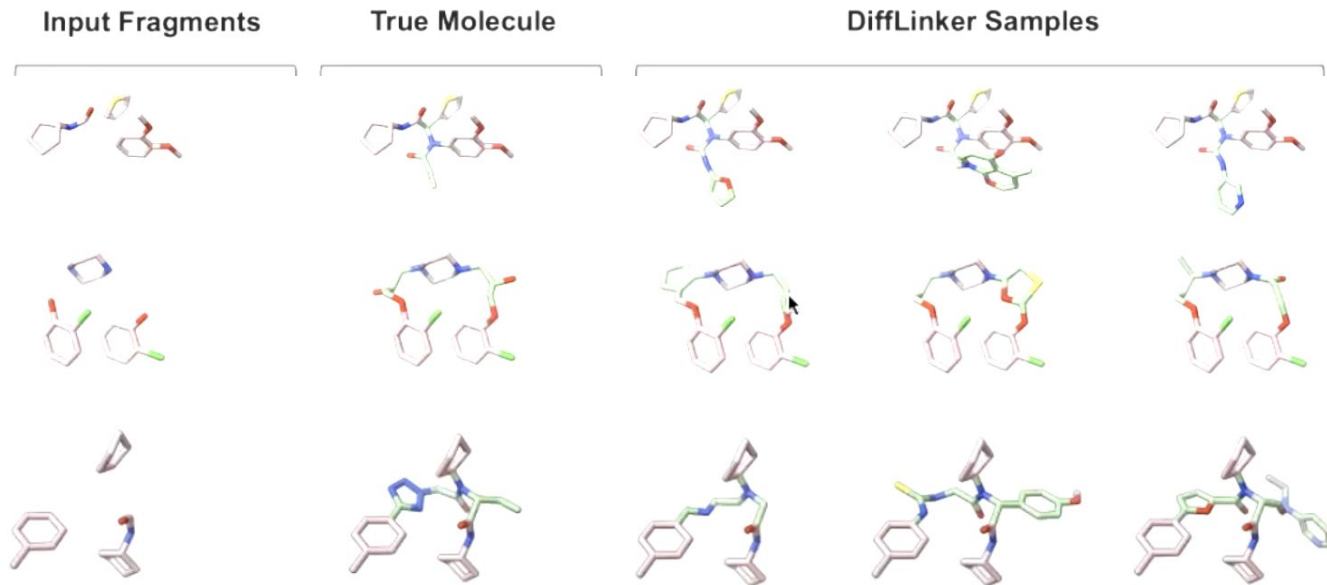
Linking Pairs of Fragments

	Method	QED $\uparrow$	SA $\downarrow$	# Rings $\uparrow$	Valid, %	Unique, %	Novel, %
ZINC	DeLinker + ConfVAE + MMFF	0.64	3.11	0.21	98.3	44.2	47.1
	3DLinker (given anchors)	0.65	3.11	0.23	99.3	29.0	41.2
	3DLinker	0.65	3.14	0.24	71.5	29.2	41.9
	DiffLinker	0.68	3.01	0.25	93.8	24.0	30.3
	DiffLinker (given anchors)	0.68	3.03	0.26	97.6	22.7	32.4
	DiffLinker (sampled size)	0.65	3.19	0.32	90.6	51.4	42.9
	DiffLinker (given anchors, sampled size)	0.65	3.24	0.36	94.8	50.9	47.7
CASF	DeLinker + ConfVAE + MMFF	0.35	4.05	0.35	95.7	51.6	55.6
	DiffLinker	0.41	4.00	0.34	85.3	40.5	41.8
	DiffLinker (given anchors)	0.40	4.03	0.38	90.2	37.3	48.4
	DiffLinker (sampled size)	0.40	4.06	0.30	63.7	60.0	49.3
	DiffLinker (given anchors, sampled size)	0.40	4.10	0.38	68.4	57.1	56.9

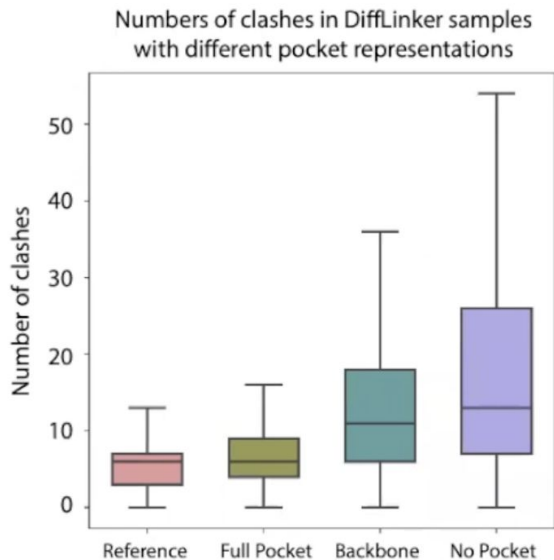
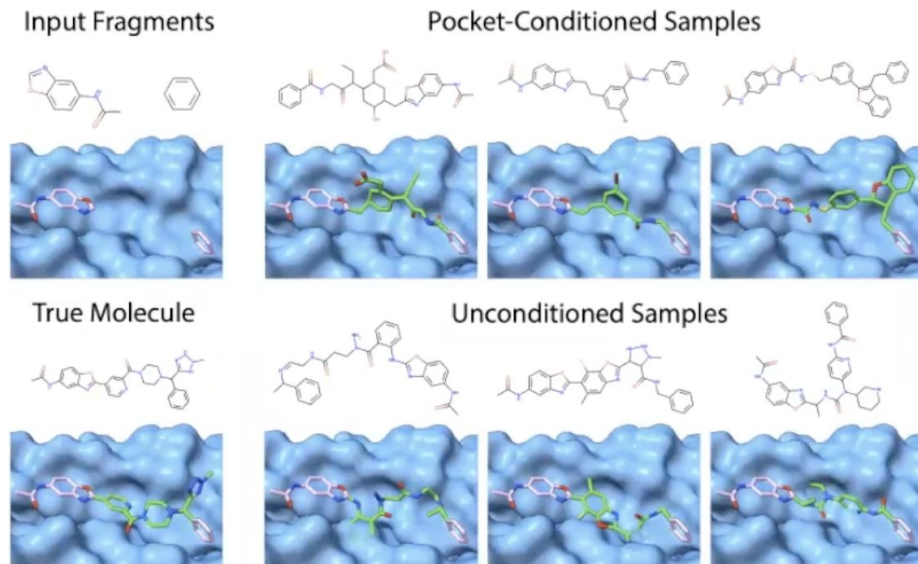


Linking >2 Fragments

Method		QED $\uparrow$	SA $\downarrow$	# Rings $\uparrow$	Valid, %	Unique, %	Novel, %
GEOM	3DLinker	0.36	3.56	0.00	16.3	73.7	—
	DiffLinker	0.48	2.99	0.75	94.0	34.7	68.6
	DiffLinker (given anchors)	0.49	3.01	0.79	94.0	35.0	68.4
	DiffLinker (sampled size)	0.45	3.27	0.75	87.2	62.4	76.0
	DiffLinker (given anchors, sampled size)	0.46	3.33	0.83	88.8	63.6	76.0



Pocket Conditioning



Method		QED $\uparrow$	SA $\downarrow$	# Rings $\uparrow$	Valid, %	Unique, %	Novel, %
Pockets	DiffLinker (pocket atoms)	0.45	3.89	1.06	88.7	62.5	73.1
	DiffLinker (pocket backbone)	0.45	3.77	0.95	90.4	60.9	72.8
	DiffLinker (unconditioned)	0.45	3.83	1.10	93.6	61.1	74.9

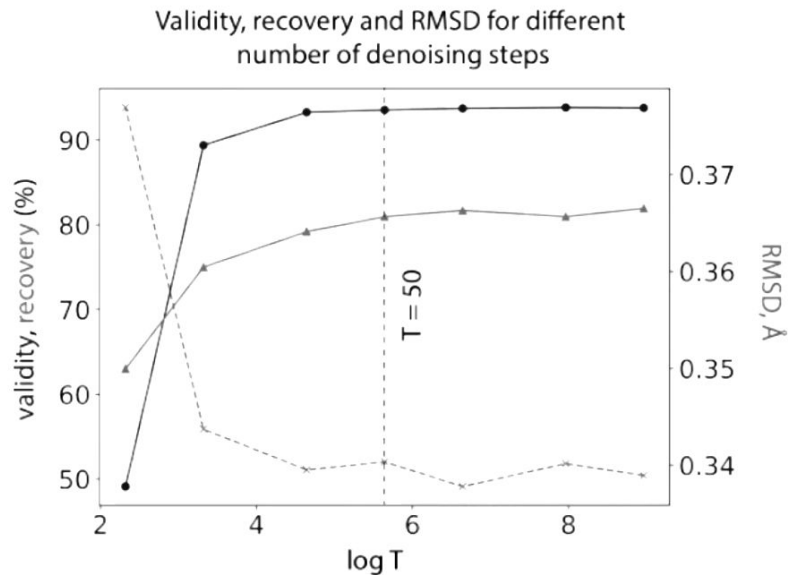
Performance

Sampling time for different datasets (with $T=500$).

Experiments were performed on a single Tesla

V100-PCI-E-32GB GPU

dataset	batch size	time per molecule, s
ZINC	128	0.37
CASF	128	0.63
Multi-Fragment	64	0.37
Pockets (full atomic representation)	32	0.46
Pockets (backbone representation)	32	0.21
Pockets (no pocket)	32	0.12



DiffLinker is robust to significant reduction of the number of denoising steps in sampling resulting in 10-fold gain in sampling speed without any performance degradation

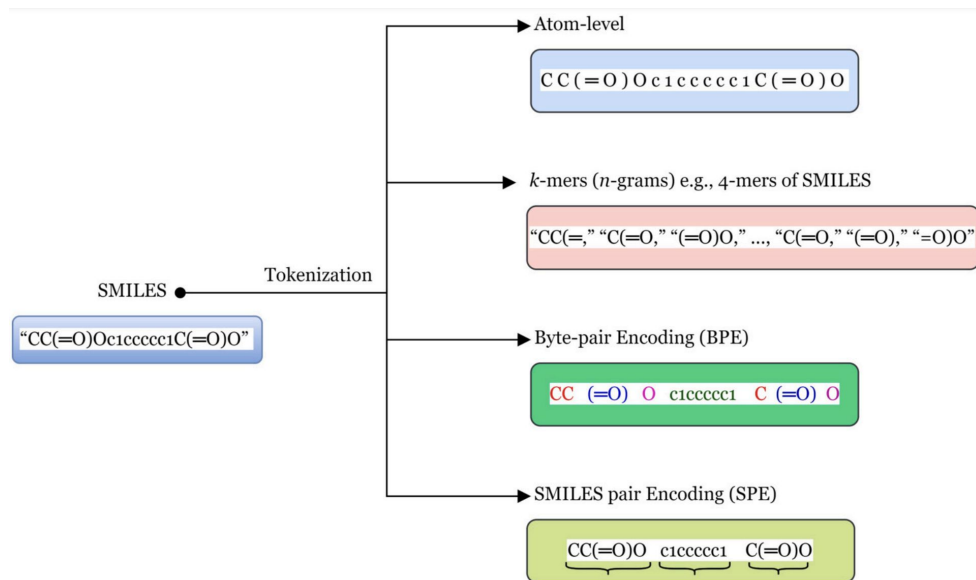
Transformer-based language models in CC

Table 1

Case studies: Transformer-Based Language Models for computational chemistry and cheminformatics.

Backbone	Encoder-based CLMs	Decoder-based CLMs	Encoder-decoder CLMs
Transformers(Attention)	BERT [28]	Guo et al. [29]	TransAntivirus (IUPAC) [30]
	MOLBERT [14]	GPT-2-based model [31]	Chemformer (BART-based) [32]
	SMILES-BERT [12]	MolXPT (GPT-2-based) [33]	MolT5 [34]
	ChemBERTa [13]	BioGPT [35]	Transformer-based models [36]
	ChemBerta-2 [37]	MolGPT (Relative Attention) [38]	X-MOL [39]
	GPT-MolBERTa [40]	Mol-Instructions [41]	SMILES transformer [42]
	RoBERTa-based model [43]		Transformer Neural Network [44]
	MolFormer [45]		Transformer-based ANN [46]
	SElFormer (SELFIES) [47]		RT (SELFIES) [48]
	Mol-BERT [49]		
	MolRoPE-BERT [50]		
	BET [51]		

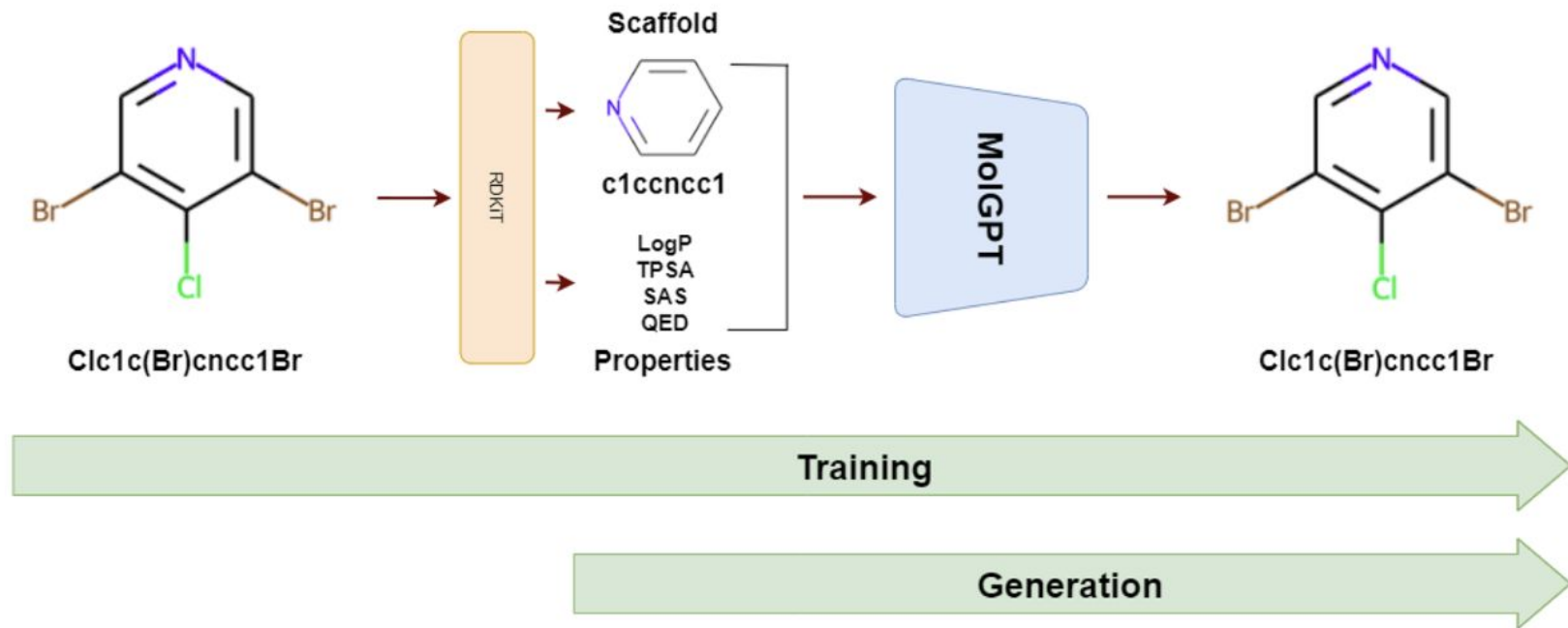
SMILES tokenization mechanisms



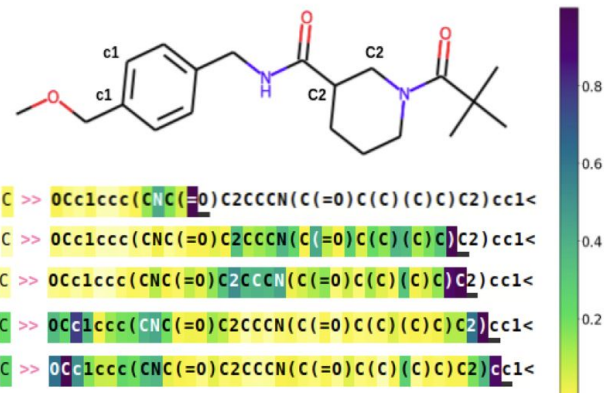
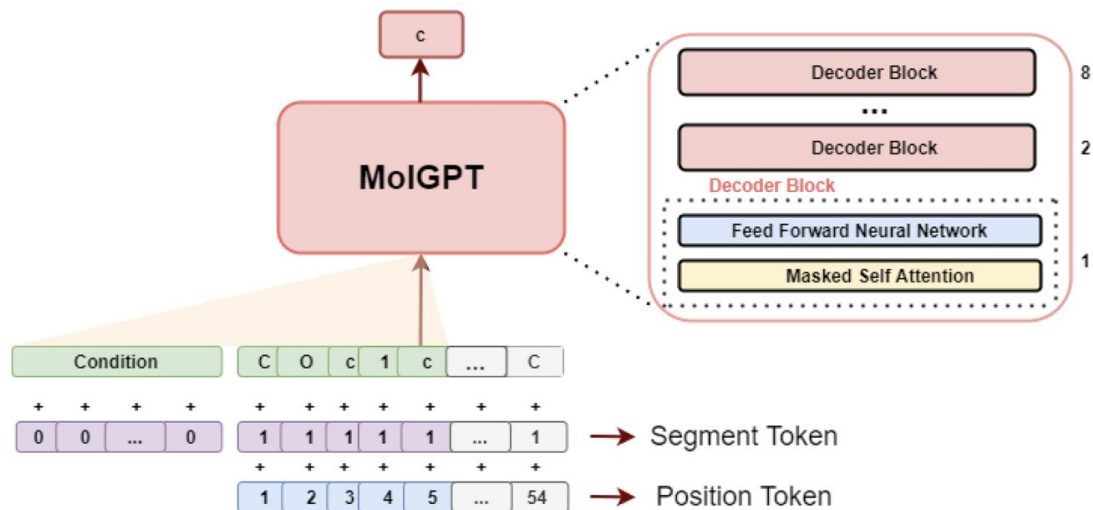
Molecular strings representations case study: ibuprofen.

Type	Molecules representation
IUPAC Name	2-[4-(2-methylpropyl)phenyl]propanoic acid
SMILES (Canonical or Isomeric)	<chem>CC(C)Cc1ccc(cc1)[C@@H](C)C(=O)O</chem>
DeepSMILES [55]	<chem>CCC)Ccccc6)) [C@@H]C)C= =O)O</chem>
SELFIES [74]	<chem>[C][C][Branch1][C][C][C][C][C] = C][C][= C][Branch1][Branch1][C][= C][Ring1][= Branch1][C@ @H1][Branch1][C][C][C][= Branch1][C][= O][O]</chem>
InChI	1S/C13H18O2/c1-9(2)8-11-4-6-12(7-5-11)10(3)13(14)15/h4-7,9-10H,8H2,1-3H3,(H,14,15)
InChIKey	HEFNNWSXXWATRW-UHFFFAOYSA-N

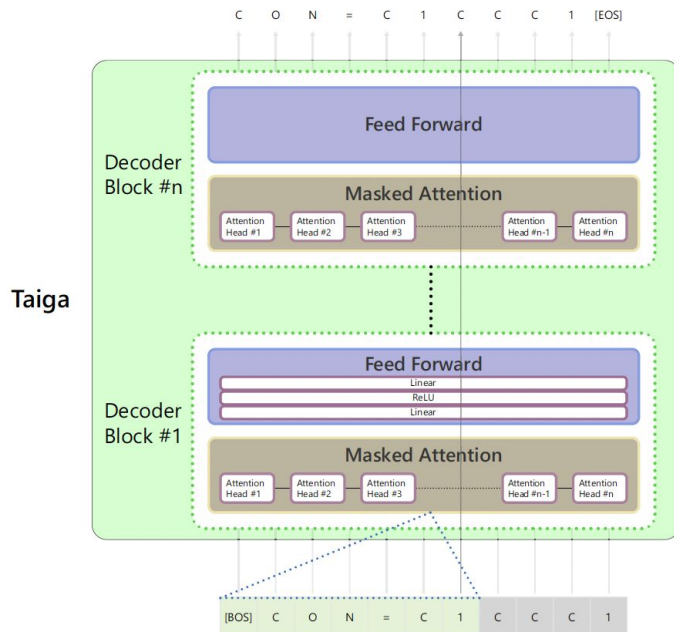
MolGPT Training



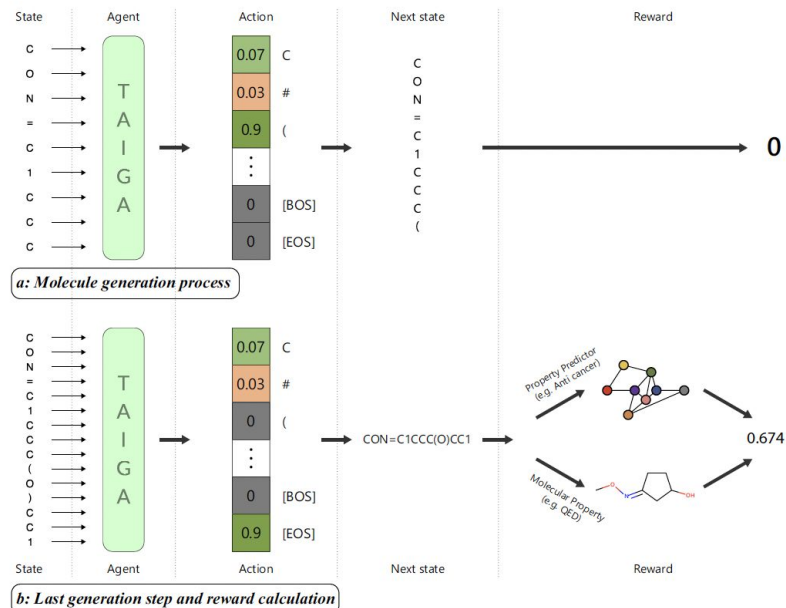
MolGPT Model Architecture



Reinforcement learning



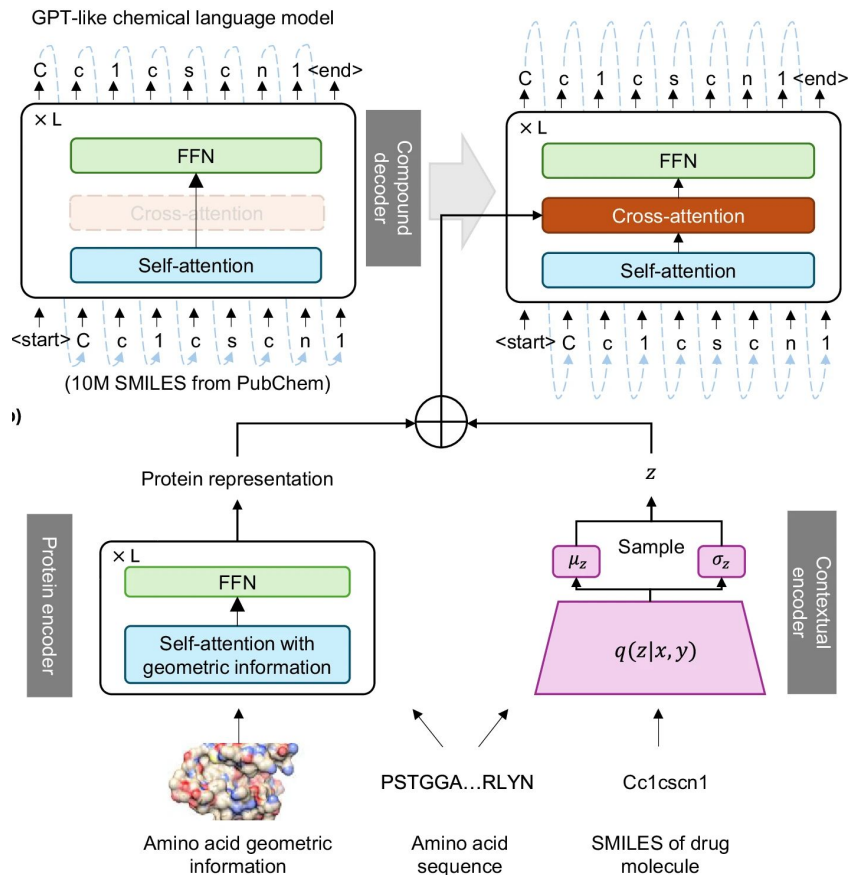
Stage 1



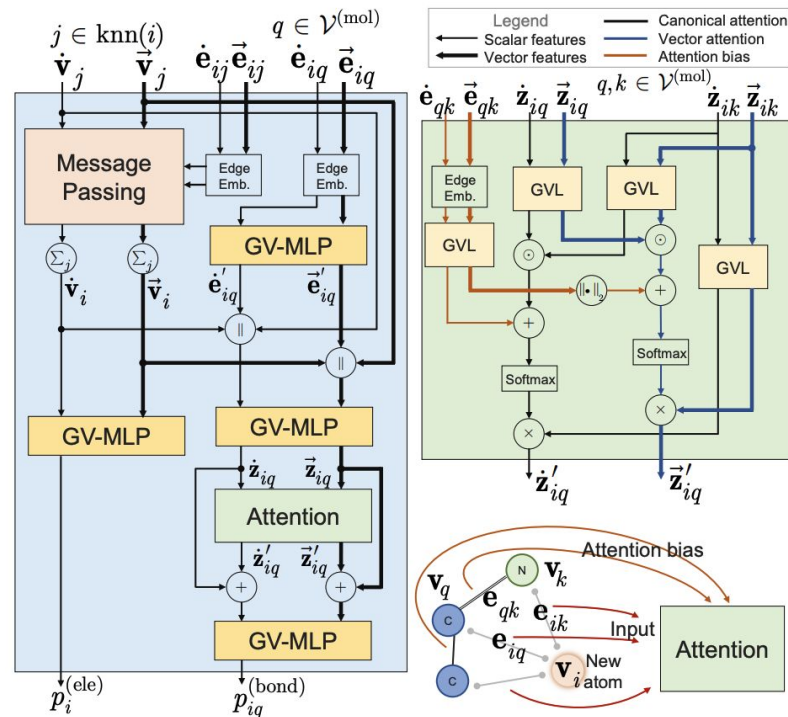
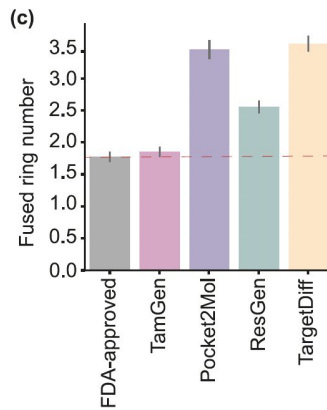
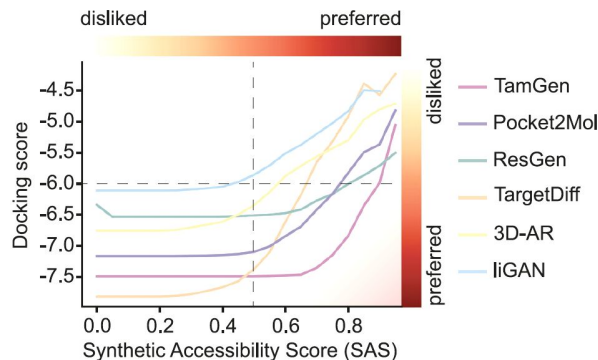
Stage 2

Unlocking Drug Discovery with Multimodal AI

- **Current challenge:** AI-generated drug candidates often have limited real-world impact due to isolated ligand-based or structure-based approaches, leading to poor outcomes in subsequent discovery stages.
- **Proposed solution:** Leveraging multimodal AI—specifically, a GPT-like chemical language model—for target-aware molecule generation and refinement.
- **Key results:** TamGen generated higher-quality, viable compounds, successfully identifying 5 promising candidates against a tuberculosis protease, with the strongest inhibitor achieving an IC_{50} of 1.9 μM .



Compound generation performance

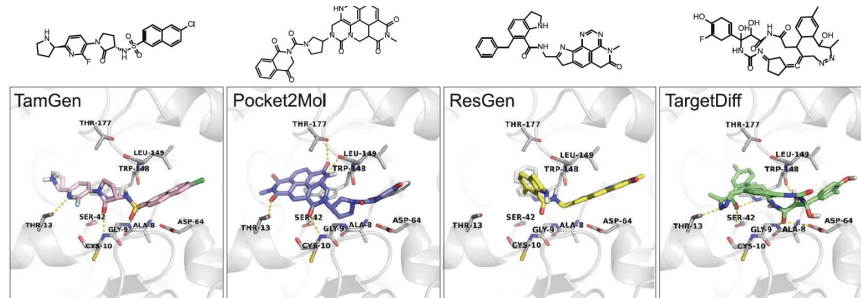
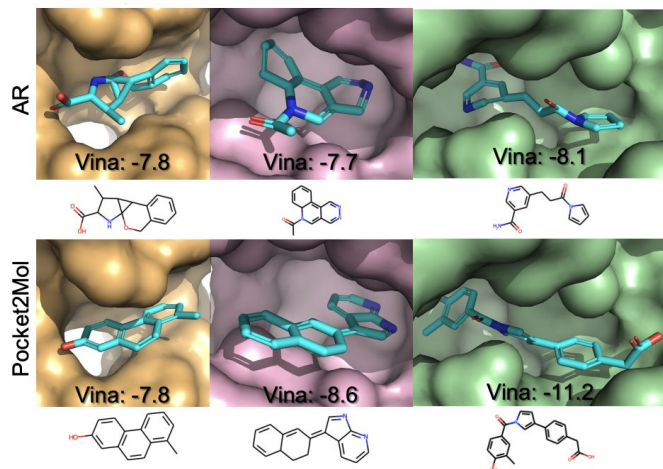


Wu et al., Nat. Comm., 2024

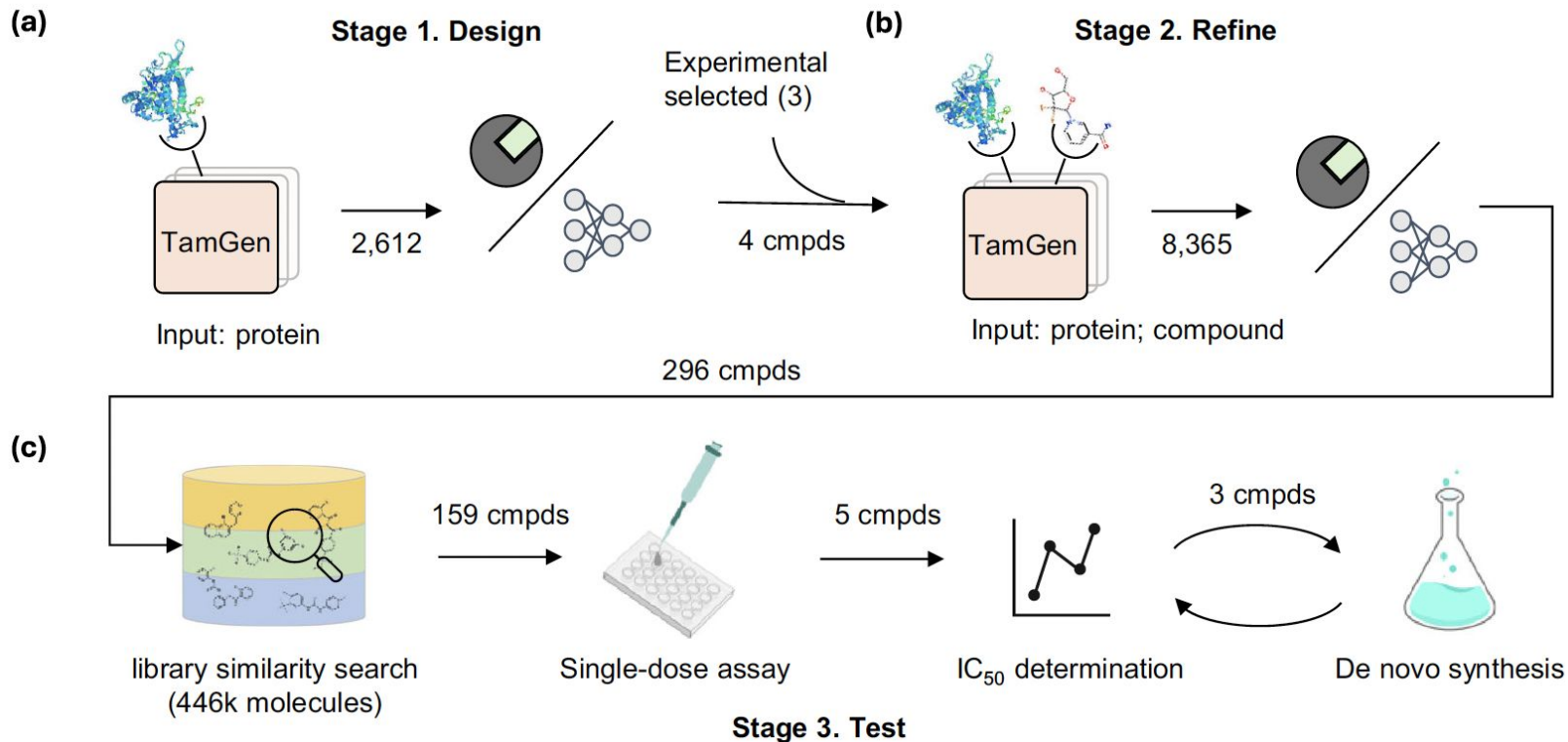
Peng et al., arXiv, 2022

Docking Visualization

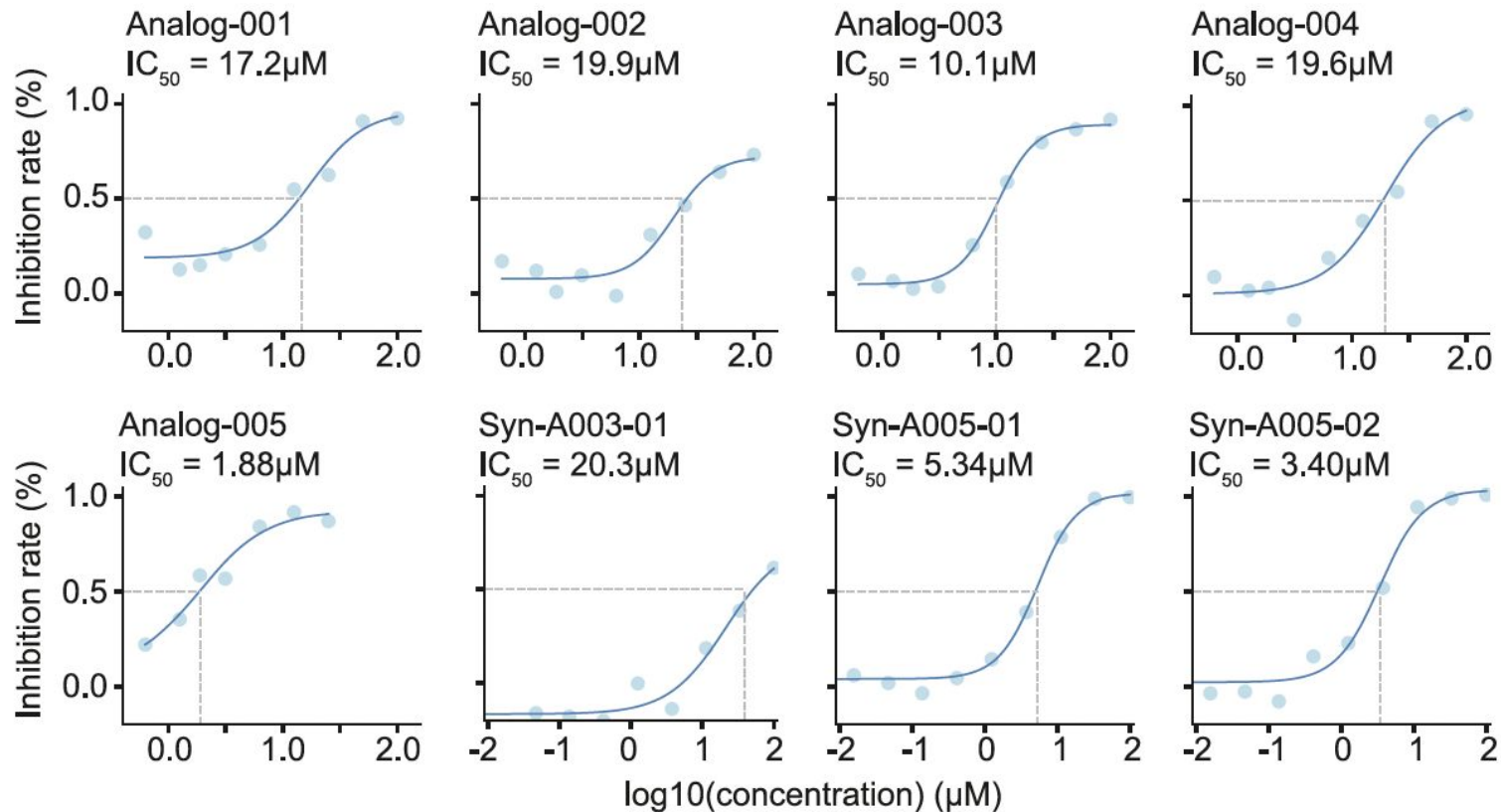
- The **protein** is the **target** — say, a viral enzyme or a cancer-related protein.
- The **ligand** is a potential drug molecule that you want to **bind to the protein** and block or activate it.
- Docking software runs simulations to explore how the **ligand can interact with the protein's binding site**.
- The docking score is a heuristic or energy-based value (usually in kcal/mol) that estimates the binding affinity



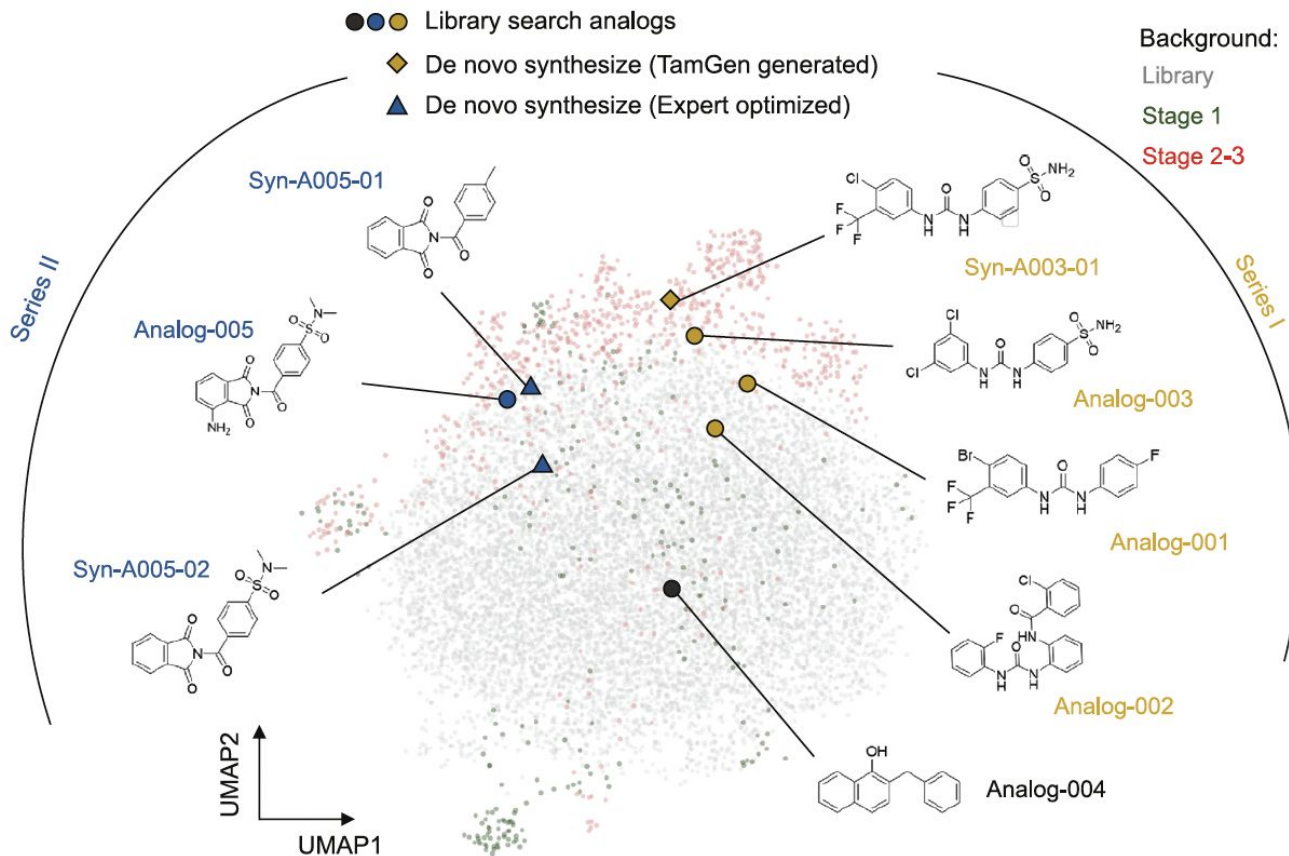
Design-Refine-Test Pipeline



Experimental validation of designed compounds

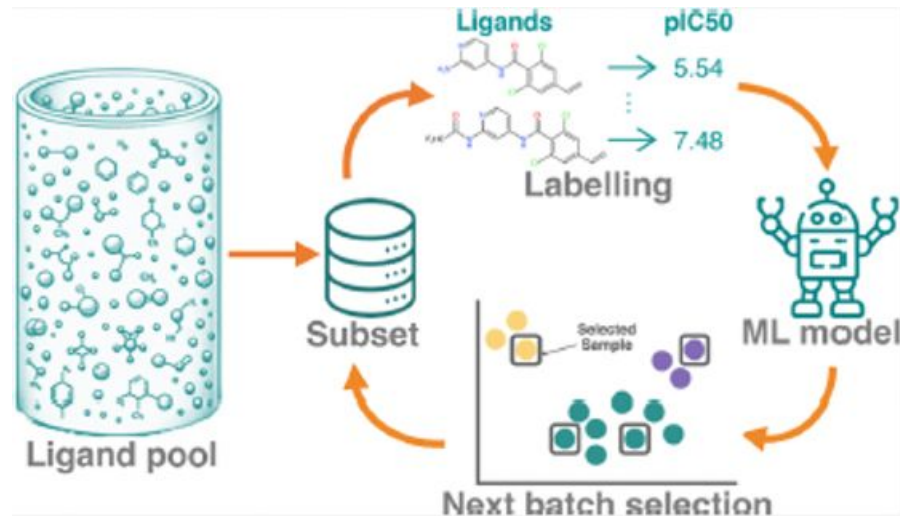


Visualization of chemical space



Active learning in drug discovery

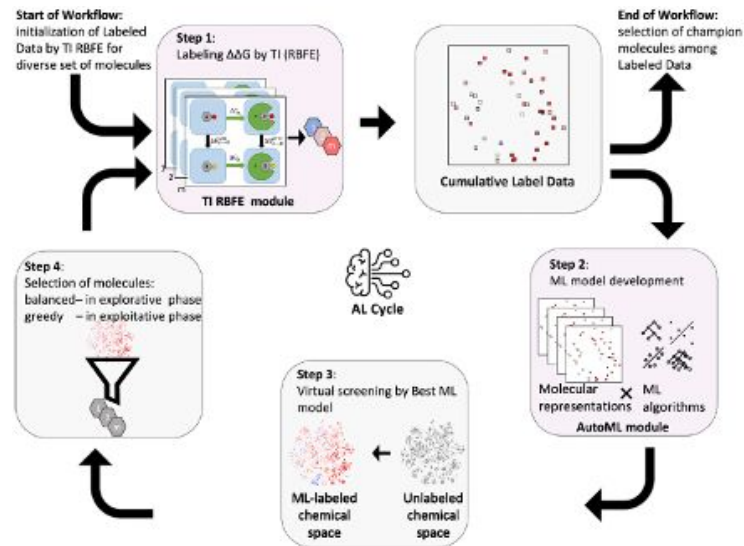
- **Active learning:** A machine learning approach focused on selective data labeling
- The algorithm queries **the most informative data points** from a pool of unlabeled data
- Reduce experimental efforts and computational resources for labeling
- Infer an accurate ML model with a **smaller, curated training dataset**
- Achieve or surpass the accuracy of models trained on larger, randomly selected datasets
- **Iteratively select data points with the highest uncertainty** or potential impact on model performance



A typical AL pipeline

In silico ligand binding free energy calculation for drug discovery

- Finding the lowest **relative binding free energy (RBFE)** among congeneric ligands
- **Thermodynamic integration (TI)** with molecular dynamics (MD) is highly effective but costly and time-intensive
- **AutoML and Active Learning (AL) integration** compounds for TI calculations are selected with ML to (1) enrich a set of molecules (exploitation), (2) improve an ML model's prediction of RBFE for an entire library
- Achieves a **100-fold increase in binding affinity** and a 20-fold speedup compared to traditional brute force methods



AI cycles of explorative and exploitative regimes

- **Exploitative** and **explorative** regimes are distinguished by **greedy selection vs balanced selection**.
- Explorative regime was used until convergence in retrospective AE.
- With AL progress, ML model's performance improves, achieving a **cross-validated MAE of 1 kcal/mol**, comparable to RBF accuracy.
- The process converged effectively, with a significant improvement in mean TI $\Delta\Delta G$ during the **exploitative AL cycle 7** compared to the random selection in cycle 6.

